



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 313: Database Management System

Batch: 23rd

Date: 05/01/2025

Mid-Term Examination

Semester: Fall 2024

Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions.

Marks: 5 x 6 = 30

1. (a) State the different types of SQL constraints. 2
(b) Write the similarities and dissimilarities between the different types of record based logical models. 4
2. (a) To create a Database Management System (DBMS), what types of information do we need? Explain. 3
(b) To design a DBMS, where does the information originated? Explain. 3
3. (a) What is meant by Questionnaires? Write the merits of Questionnaires to determine the requirements of a system. 3
(b) Briefly explain the Types of Interviews and Questionnaires to determine the requirements of a system. 3
4. (a) What is meant by prototyping of a system? 3
(b) Briefly explain the useful and drawbacks of prototyping. 3
5. (a) Write the merits and demerits of an interview to collect the requirements of a DBMS. 3
(b) Briefly explain the different types of feasibility study with the steps in feasibility analysis. 3
6. Design an E-R Diagram and write the SQL command to create all the relations for the following scenario: 6

Employee	Department	Project	Dependent
Employee_ID	Dept_No	Proj_ID	Depn_Name
Employee_Name	Dept_Name	Proj_Name	Depn_Gender
Employee_Gender	Dept_Location	Proj_Location	Depn_relation
Employee_Address			
Employee_DoB			
Employee_DoJ			

7. Write the Data Flow Diagramming rules. 6

Source / Data Flow

Source Skills



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 317: Computer Networks

Batch: 23rd
Date: 01/01/2025

Semester: Fall 2024
Time: 1 Hour 30 Minutes
Total Marks: 30

Mid-Term Examination

Answer **any five** questions of the followings.

5 × 6 = 30

1. (a) Define **Computer Networks**. Briefly explain **fault tolerance** and **scalability** of computer networks. 3
- (b) Write the differences between the **Bandwidth** and the **Speed** of computer networks. 3
2. (a) Define **Throughput** with advantages and disadvantages. 3
- (b) What is **Repeater**? Write the differences between **HUB** and **Switch**. 3
3. (a) Why a subnet mask is necessary in IP addressing? Identify the class of the following IP addresses: 2
 - i. 10100001.11110000.11001100.00001111
 - ii. BB.AC.E2.0
- (b) Consider the IP address block, 22.204.134.99/20. **Subnet** the IP block using a fixed length subnet masks (FLSM). Answer the following: 2
 - i. What is the **fifth** IP address in block 5?
 - ii. What is the **3rd** **usable** IP address in block 3?
 - iii. In which block is the given IP address **22.204.134.99** located?
 - iv. Write the subnet mask of the FLSM.
4. (a) Consider the subnet mask of a network X.X.X.11000000. Here, X indicates 8 bits of 1. How many valid hosts can be accommodated using this subnet? 2
- (b) Suppose you are accessing a local network as Host A. You want to connect to Host B using a layer 2 networking device. Assume for the first time Host A and B connect with each other. Which networking protocol is used for this connection? Describe all the steps for obtaining the MAC address of the destination host. 4

5. (a) People use the Internet for different purposes. Using internet they chat with each other, they doing mathematical problems, view images, watching and listening audio and video files. Which data formats they are accessing in this scenario and how computer represent them? 3
- (b) Write the differences between Circuit Switching and Packet Switching. 3
6. (a) What is Loopback Address? What is equal Classless Inter-Domain Routing (CIDR) value of 255.255.248.0 subnet mask? 2
- (b) Consider you are given the address space 192.16.0.0/20. Now Write the answers of the following questions according to the host requirements for different departments that are given in the following table: 4

Department	No. of the hosts required
Sales 1	1000
Service	800
Accounts	20
Sales 2	1200
HR	600

- What is the 5th IP address of service department?
 - What is the subnet mask of HR department?
 - What is the 5th usable IP address of Sales 1 department?
 - What is the host bit of Accounts department?
7. (a) What are the purposes of Dynamic Host Configuration Protocol (DHCP) relay agent? 2
- (b) Explain the messages of DHCP protocol. 4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE311: Computer Architecture
Mid-Term Examination

Batch: 23rd

Date: 14/01/2025

Semester: Fall 2024

Time: 1 Hour 30 minutes

Answer any **five** questions of the following.

$5 \times 6 = 30$

1. a) Describe the impact of reduced cycles per instruction (CPI) and reduced clock cycle time on CPU efficiency. 2
b) In an IAS computer, if the instruction 'ADD X' is given, explain each step of the CPU clock cycle and the state of the corresponding registers during each step. Assume $X = 200$, $M(200) = 20$, and $AC = 30$. 4
2. a) What is interrupt? 1
b) State the differences between synchronous and asynchronous bus. 2
c) What are I/O devices? Briefly explain their types with examples and applications. 3
3. a) Given a main memory of 32K words (each 12 bits) and a cache memory of 512 words (each 12 bits), determine the minimum CPU address size required to access any location in the main memory. 2

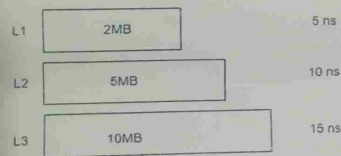


Figure 1

- b) Given a 1KB data block residing in the L3 cache, determine the access time during the initial access. 2
- c) Assuming the same 1KB data block is accessed again subsequently and the CPU utilizes an LRU cache replacement policy then calculate the expected access time for the second access. Also determine how much time is saved this time. 2
4. a) Briefly describe the classification of memory. 2
b) What are analysis and synthesis, and how are they different from each other? 2
c) How many types of register level components are there? Define and explain. 2
5. a) State the hardware milestones in all generations of computers. 2
b) Draw and explain internal organization of a CPU and cache memory. 4
6. a) What is microcontroller? 1
b) State and explain the major control states of CPU in third generation computers. 2
c) "Batch processing played a significant role in the early development of operating systems"-Justify the statement. 3

7. a) What is a data hazard? Compare the results of the program below in both scenarios: one where sequential execution is used and another where Instruction 1 is executed in Pipeline 1 while Instruction 2 is executed in Pipeline 2. Also explain if any data hazard occurs here.

```
#include <stdio.h>
int main() {
    int a = 5, b = 10, c = 5;
    c = a + b; //Instruction 1
    b = c * 2; //Instruction 2
    printf("Value of cn:%d\n", c);
    printf("Value of b: %d\n", b);
}
```

- b) Suppose there is a program with 1000 instructions, where 80% of the instructions can be executed using pipelining. The pipeline has a depth of 6 stages, and the clock cycle time is 10^{-9} seconds. How much time will the CPU take to execute the entire program?

9/1/25



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE315: Microprocessors & Microcontrollers

Batch: 23th

Date: 07/01/2025

Midterm Examination

Semester: Fall 2024

Time: 1 Hour 30 minutes

Total Marks: 30

Answer any **five** questions of the following.

$5 \times 6 = 30$

1. (a) Compare Microprocessor and Microcontroller based on their architecture and applications. 3
(b) Draw and explain the block diagram of 8086 Architecture showing BIU and EU components. 3
2. (a) What are the criteria for choosing a microcontroller for a specific application? 2
(b) Explain the role of segment registers in 8086. How is physical address calculated using segment registers? 4
3. (a) List and explain any four addressing modes of 8086 with suitable examples. 3
(b) What are the different types of registers in 8086? Explain the function of general-purpose registers in detail. 3
4. (a) Describe the function of the following registers in 8086. 3
i. Stack Pointer
ii. Base Pointer
iii. Instruction Pointer
(b) Draw and explain the Flag register of 8086 microprocessor. What is the significance of each flag? 3
5. (a) Explain the instruction format of 8086 microprocessor. 2
(b) Generate the machine code for the following instruction: 4
MOV BX, [ABCDH]
6. (a) What are the advantages and disadvantages of RISC and CISC architecture? 3
(b) Explain Von Neumann and Harvard architectures with diagrams. 3
7. (a) What is the function of the CWCR and MRSR registers in DMA? 2
(b) What is DMA? Explain its working principle with diagram. 4



HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT317

Mid-Term Examination

Date: 1/12/2025

Course Title: Engineering Mathematics

Semester: Fall 2024

Total Time: 1 Hour 30 minutes

Total Marks: 30

Answer any three (03) out of the following questions:

$3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

- | | Marks |
|---|-------|
| Q1. (a) Define the Laplace Transformation. What is its significance. | [2] |
| (b) Find the Laplace transformation of $4\sin^2 t$ | [3] |
| (c) Obtain the Laplace transformation of | [5] |
| (i) $t^2 e^{2t} \sin 4t$ | |
| (ii) $t^2 \sin 4t$ | |
| Q2. (a) Using Laplace transformation, evaluate $\int_0^\infty \frac{e^{-t} \sin t}{t} dt$ | [3] |
| (b) Using Laplace transformation, Show that, $\int_0^\infty \frac{e^{-t} - e^{-3t}}{t} dt = \ln 3$ | [3] |
| (c) Find the inverse Laplace transform: | [4] |
| i. $L^{-1}\left\{\frac{1}{s(s^2+1)}\right\}$ | |
| ii. $L^{-1}\left\{\frac{1}{s^3+4s^2+3s}\right\}$ | |
| Q3. (a) State and prove the convolution theorem of Laplace transforms. | [5] |
| (b) Solve the differential equation $\frac{d^2 y}{dt^2} - 2\frac{dy}{dt} - 8y = 0$ | [5] |
| using Laplace transforms, given $y(0) = 3$ and $y'(0) = 6$. | |
| Q4. (a) Define even and odd function with proper example. | [2] |
| (b) Define the Fourier series and find Fourier coefficients. | [3] |
| (c) Determine the Fourier series of the function $f(x) = \begin{cases} 0, & -\pi \leq x \leq 0 \\ 1, & 0 \leq x \leq \pi \end{cases}$ | [5] |
| Q5. (a) Find the Fourier series of $f(x) = x$ over the interval $[0, \pi]$ | [4] |
| (b) Find the Fourier series of the function | [6] |
| $f(x) = \begin{cases} -1, & -\pi \leq x \leq -\frac{\pi}{2} \\ 0, & -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \\ 1, & \frac{\pi}{2} \leq x \leq \pi \end{cases}$ | |